

QQ: A Language Metadata Toolkit for Multilingual NLP

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One language, many identifiers

German

OPUS
de

WMT
deu

Legacy data
ger

FineWeb-2
deu_Latn

Glottolog
stan1295

One language, many identifiers

German

OPUS	WMT	Legacy data	FineWeb-2	Glottolog
de	deu	ger	deu_Latn	stan1295

many identifiers

One language, many identifiers

German

OPUS	WMT	Legacy data	FineWeb-2	Glottolog
de	deu	ger	deu_Latn	stan1295

many identifiers

×

many languages

- Development started two years ago
- Solve my own annoyances
- Qwanqwa (ቋንቋ) means **language** in Amharic

QQ overview

1 Metadata sources

Resource 1

de deu

Resource 2

deu stan1295

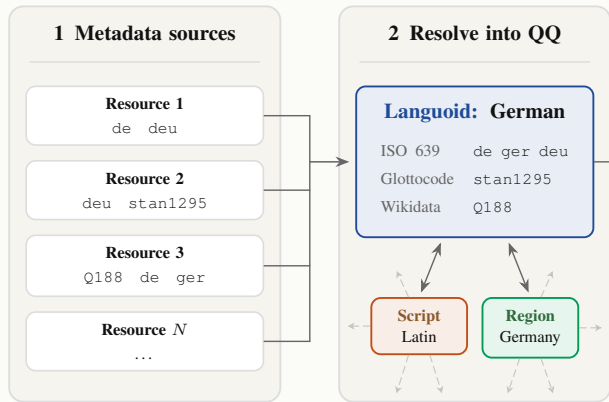
Resource 3

Q188 de ger

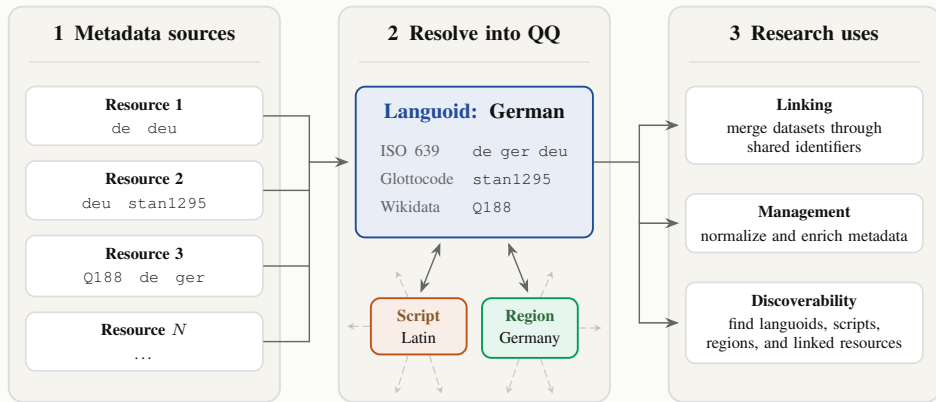
Resource N

...

QQ overview



QQ overview



Merging overlapping records

Source 1
deu
--
ger
--

Source 2
--
--
ger
de

Source 3
deu
stan1295
--
--

Source 4
deu
--
--
--

Merging overlapping records

Source 1
deu
--
ger
--

Source 2
--
--
ger
de

Source 3
deu
stan1295
--
--

Source 4
deu
--
--
--

Merging overlapping records

Source 1
deu
--
ger
--

Source 2
--
--
ger
de

Source 3
deu
stan1295
--
--

Source 4
deu
--
--
--

Merged record
de
deu
ger
stan1295

Merging overlapping records

Merged record
de
deu
ger
stan1295

Metadata
Names
Identifiers
Language family
Scripts & regions
Speaker estimates
Endangerment status
Linked resources
...

QQ interfaces

Interfaces

- **Python package:** `pip install qwanqwa`
- **Browser explorer:** `wesselpoelman.nl/qq`
- **CLI:** `qq get nld`

Python: identifiers

```
from qq import Database

db = Database.load()

dutch = db.get("nl")
dutch.name # "Dutch"
dutch.iso_639_3 # "nld"
dutch.speaker_count # 24085200

dutch2 = db.guess("nld")
dutch3 = db.guess("dutc1256")
dutch4 = db.guess("dut-Latn-NL")

assert dutch == dutch2 == dutch3 == dutch4
# True
```

Python: conversion

```
db.convert("nl-Latn-BE", IdType.ISO_639_3)
# "nld"
db.convert("nld", IdType.GLOTTOCODE)
# "dutcl256"

db.convert("nl", IdType.ISO_639_3)
# "nld"
db.convert("dutcl256", IdType.ISO_639_3)
# "nld"
```

Python: traversal

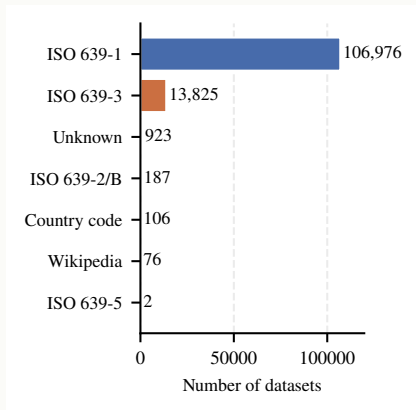
```
dutch.parent
# "Global Dutch"
dutch.parent.parent
# "Modern Dutch"
dutch.family_tree
# ["Global Dutch", ..., "Germanic", ..., "Indo-European"]
dutch.scripts
# [Script(Latin, code=Latn)]
dutch.regions
# ["Aruba", "Belgium", ..., "Netherlands", "Suriname"]
dutch.name_in("fr")
# "Néerlandais"
dutch.name_in(db.get("fr"))
# "Néerlandais"
dutch.endonym
# "Nederlands"
```


Explorer: metadata and graph

Live Demo!

if it works...

Case study: HuggingFace Hub



98.9%

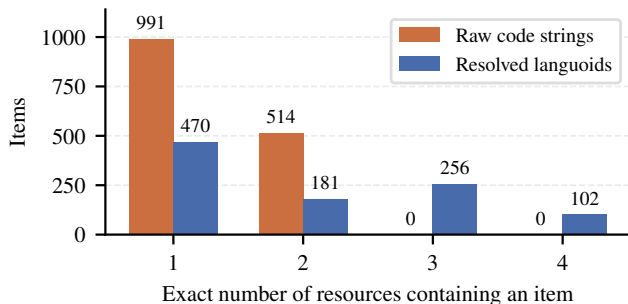
resolved

8,144 / 8,237 unique tags

7,943

unique languoids

Case study: linking resources



	Concept./WN	Etymon	Phonot.	QQ
Dutch	NL	nld	dutc1256	Dutch
German	DE	deu	stan1295	German

Acknowledgements

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Try QQ

- **pip install qwanqwa**
- Active development
- Feedback welcome:
`wessel.poelman@kuleuven.be`



Explorer: languoid

LANGUOID

Dutch

nl / nld / dutc1256

NAMES

Name

Dutch

Copy

Endonym

Nederlands

Copy

Description

West Germanic language.

Copy

IDENTIFIERS

BCP-47

nl

Copy

ISO 639-1

nl

Copy

ISO 639-2B

dut

Copy

Graph

ViewMixed

```
graph TD; Dutch((Dutch)) --- GlobalDutch(Global Dutch); Dutch --- AntwerpBrabants(Antwerp Brabants); Dutch --- Antwerps(Antwerps); Dutch --- Bildts(Bildts); Dutch --- Plus25((+25)); Dutch --- Latin(Latin); Dutch --- Netherlands(Netherlands); Dutch --- Belgium(Belgium); Dutch --- Plus5((+5));
```

Resources

Datasets

Explorer: script

SCRIPT

Latin

Latn

NAMES

Name	Latin	Copy
Example	Di	Copy

IDENTIFIERS

ISO 15924	Latn	Copy
Canonical ID	script:latn	Copy

UNICODE

Unicode script	Latin	Copy
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Graph View Mixed ▾

```
graph TD; Ani["//Ani"] --- Latin["Latin"]; Gana["//Gana"] --- Latin; Areare["Are'are"] --- Latin; Auhelawa["Auhelawa"] --- Latin; Plus6632["+6632"] --- Latin;
```

Resources

No external resources found.

Explorer: region

REGION

Belgium

BE

NAMES

Name	Belgium	Copy
Official name		Empty

IDENTIFIERS

Country code	BE	Copy
Canonical ID	region:be	Copy

Graph View Mixed ▾

```
graph TD; Belgium --- Dutch; Belgium --- French; Belgium --- German; Belgium --- Picard; Belgium --- Plus3[+3]; Belgium --- Duployan; Belgium --- Hebrew; Belgium --- Latin;
```

The graph illustrates the linguistic and regional connections of Belgium. The central node 'Belgium' is connected to eight peripheral nodes: 'Dutch', 'French', 'German', 'Picard', '+3' (indicating three more languages), 'Duployan', 'Hebrew', and 'Latin'. The nodes are color-coded: blue for 'Dutch', 'French', 'German', and 'Picard'; orange for 'Latin', 'Hebrew', and 'Duployan'; and a dashed blue box for '+3'.

Resources